



Large Language Models in Data Science

Week 4: Text Classification — Intent Routing for AMU Chatbot

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Session Overview

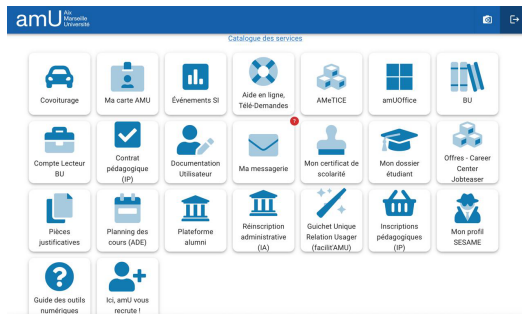
Lecture (15min)

1. Problem framing: intent classification
2. Dataset design and labels
3. Baselines: regex rules
4. Embeddings + Logistic Regression
5. Zero-shot NLI pipeline
6. LLM prompting for classification
7. Comparison and trade-offs

Lab (2.5h)

- ▶ Build 4 classifiers (regex, embed+LR, zero-shot, LLM)
- ▶ Measure accuracy and latency
- ▶ Analyze failure cases and label phrasing
- ▶ Extend training data and re-evaluate
- ▶ Optional: add a new intent

AMU Chatbot



- ▶ AMU is developing a student support chatbot (in French and English).
- ▶ The chatbot needs to route student queries to the right AMU service/tool.
- ▶ Reliable intent classification is critical for user satisfaction.

Project: AMU Chatbot Brain

- ▶ **Goal:** Map student queries to AMU services (*intent routing*).
- ▶ **Task:** Given a short query, predict one of 5 intents.
- ▶ **Why it matters:** Enables reliable hand-off to the right tool/API.

Intents (lab setup)

- ▶ `get_schedule` (Planning des cours / ADE)
- ▶ `check_email` (Ma messagerie)
- ▶ `register_classes` (Inscriptions pédagogiques / IP)
- ▶ `get_student_card` (Ma carte AMU)
- ▶ `find_library_info` (BU / Compte Lecteur BU)

Four Approaches at a Glance

- ▶ **Regex baseline**: Fastest and transparent; brittle, hard to scale.
- ▶ **Embeddings + LogisticRegression**: Train on labelled examples; measure held-out accuracy and inference time.
- ▶ **Zero-shot classification (MNLI)**: No training; label phrasing matters; moderate latency.
- ▶ **LLM prompt**: Describe labels in the prompt; measure accuracy, latency, and invalid outputs.

Evaluation Plan

- ▶ **Metrics:** *Accuracy* (correct label) and *Latency* (ms per query).
- ▶ **Protocol:** Run all methods on the same test set; record predictions+timings.
- ▶ **Compute note:** Latency varies by device (CPU vs. GPU); compare on same hardware when possible.
- ▶ **Reproducibility:** Pin model names/revisions; save prompts and candidates; fix seeds where relevant.

Design Tips

- ▶ **Regex:** Prefer precise patterns; avoid overbroad keywords (e.g., plain "amu").
- ▶ **Embeddings:** Use multilingual sentence models; validate with cross-validation if data grows.
- ▶ **Zero-shot:** Use descriptive, possibly multilingual labels (*emploi du temps* vs. *get_schedule*).
- ▶ **LLM prompting:** Specify role, label set, and *strict output format* (JSON / one token).

Key Takeaways & Recommendations

- ▶ Start with **embeddings + LR** for a strong, fast baseline.
- ▶ Use **zero-shot** for quick prototyping or new intents without data.
- ▶ Keep a small **regex layer** for guardrails and trivial routes.
- ▶ Add an **LLM fallback** for ambiguous/edge cases; require structured outputs and log decisions.